

SDS Exercise Sheet 2: Dialogue System Toolkit and Corpora

Nurul Lubis

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1 Introduction

In this exercise you are going to take a closer look at a toolkit designed for implementing dialogue system modules and pipeline called ConvLab3 [2].

The objective of this exercise are as follows:

- Understand the general setting of a statistical spoken dialogue system,
- Gain intuition on how a dialogue corpus look like and how they can be used to train machine learning models,
- Set up a simple dialogue system pipeline and interact with it.

2 ConvLab3

ConvLab3 is a dialogue system toolkit for task-oriented dialogue (TOD) systems. It provides an array of standard and state-of-the-art models covering all modules in a TOD system, i.e. natural language understanding (NLU), dialogue state tracking (DST), dialogue policy, user simulator policy, natural language generation (NLG), and end-to-end models. This allows straightforward plug-and-play experimentation when developing a specific module. Furthermore, the toolkit provides a unified format interface for accessing a variety of dialogue system corpora. More details on the toolkit can be found in [2].

2.1 Dialogue Corpora

ConvLab3 operates with the unified data format of dialogue corpora. A unified data format serves as the adapter between TOD datasets and models: datasets are first transformed to the unified format and then loaded by models. A unified dataset in ConvLab-3 consists of dialogues, ontology, and database (or API interface). The unified format is defined over user goal, dialogue acts, state, API result, etc. to support common dialogue modeling tasks, while keeping other dataset-specific annotation (e.g., emotion) intact. Similarly, a model supporting the unified format can access all transformed datasets.

In this and the following exercises, we will be working on **Simple MultiWOZ 2.1**, which is an abbreviated version of MultiWOZ corpus [1], containing only hotel and restaurant domains. However, the repository contains other dialogue corpora which you can freely analyze and play around with.

2.2 Dialogue System Modules

A modular dialogue system agent can be comprised of a wide range of modules. ConvLab-3 fully supports the building of any combination and allows easy substitution of components into one dialogue system pipeline. To support this functionality, each module implements a `predict` function with a standardized input and output formats.

The integration of the modules into a dialogue pipeline is done by the `PipelineAgent` class in `convlab/dialog_agent/agent.py`. In particular, the `response` function runs this pipeline in order to generate the next system response.

3 Exercises

We will explore the structure of the toolkit and start with a set of baseline modules to create our first dialogue system pipeline.

3.1 Codebase

Throughout the course you will maintain a personal repository of your implementations. Fork the following base project from git:

<https://git.hhu.de/sds/convlab3>

Set your repository to be **private**. To ensure that your repository is visible for grading, add user `lubis` as a project member.

3.2 Pre-trained model

We provide small pre-trained NLU and policy models which can be downloaded from sciebo (link and password in ILIAS).

3.3 Tasks

1. Explore the code and dataset structures of the ConvLab3 toolkit.
2. Complete the implementation of a simple dialogue system pipeline based on the code in `convlab3/examples/agent_examples/interact.py`. Use the pre-trained models provided above as the appropriate modules.

3.4 Results and Discussion

In your final report, write a short report covering, but not limited to, the following points. Additional insights, analysis, and discussion are highly encouraged. Push your completed `interact.py` into your git repository and put the link to your repository in the final report.

- Based on the unified format, describe the properties of an arbitrary dialogue corpus. What components and labels are present? How could they be utilized with machine learning to build a dialogue system?
- Try interacting with the dialogue system, posing as the user. Provide a dialogue example from this interaction. What are the weaknesses of this dialogue system? How can they be improved?

References

- [1] BUDZIANOWSKI, P., WEN, T.-H., TSENG, B.-H., CASANUEVA, I., STEFAN, U., OSMAN, R., AND GAŠIĆ, M. Multiwoz - a large-scale multi-domain wizard-of-oz dataset for task-oriented dialogue modelling. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (2018).
- [2] ZHU, Q., GEISHAUSER, C., LIN, H.-C., VAN NIEKERK, C., PENG, B., ZHANG, Z., FENG, S., HECK, M., LUBIS, N., WAN, D., ZHU, X., GAO, J., GASIC, M., AND HUANG, M. ConvLab-3: A flexible dialogue system toolkit based on a unified data format. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (Singapore, Dec. 2023), Y. Feng and E. Lefever, Eds., Association for Computational Linguistics, pp. 106–123.